

Removable and Essential C^2 Singularities on Punctured Disks and a Reduction of the C^2 Isometric Immersion Problem

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Abstract

We classify C^2 Riemannian singularities on punctured disks into removable and essential types, using purely intrinsic criteria. This yields a reduction of the C^2 isometric immersion problem: metrics with only removable singularities admit no intrinsic obstructions to immersion in $\mathbb{R}^3$, while essential singularities—characterized by unbounded curvature, non-Euclidean circle collapse, or irregular polar expansions—are the sole source of potential obstructions. This decomposition separates the problem into a tractable, geometrically natural regime and a small pathological class, clarifying the landscape for future research.

1 Introduction

The classical C^2 isometric immersion problem asks:

Given a two-dimensional C^2 Riemannian manifold (M, g) , does there exist a C^2 isometric immersion (or embedding) of (M, g) into $\mathbb{R}^3$?

This question is at the heart of low-regularity embedding theory. It lies between the highly flexible C^1 Nash–Kuiper theorem and the smooth or high-codimension existence theorems of Nash and Günther, and has remained open in its full generality.

Most of the classical work on isometric embeddings and immersions is *extrinsic*: it studies embeddings of a *given* smooth manifold into $\mathbb{R}^3$ or higher-dimensional Euclidean spaces, often under curvature or convexity assumptions (as in the Weyl problem for positively curved metrics on S^2). By contrast, the intrinsic geometry of possible singularities of C^2 metrics—and their role in the immersion problem—has received far less structural attention.

The purpose of this note is to isolate a clean intrinsic dichotomy for C^2 metrics on punctured disks, and to explain how this leads to a reduction of the C^2 isometric immersion problem. Concretely, we:

- Define and characterize *removable* and *essential* C^2 metric singularities on a punctured disk $D^2 \setminus \{0\}$ in purely intrinsic terms.
- Show that removable singularities are exactly those for which the metric completion adds a genuine C^2 Riemannian point, with bounded curvature and Euclidean first-order behavior of small geodesic circles.
- Observe that metrics whose incomplete ends are all removable admit no *intrinsic* obstruction to C^2 isometric immersion into $\mathbb{R}^3$; any difficulty must arise from extrinsic compatibility (Gauss–Codazzi) or from essential singularities.

- Point out that the classical Weyl class of C^2 , positively curved sphere metrics lies entirely within the removable regime, so Weyl-type methods, while powerful in that restricted setting, do not touch the intrinsically delicate cases that remain for the C^2 immersion problem.

The resulting picture is that the C^2 immersion problem naturally splits into a geometrically benign, “intrinsically regular” regime and a small but subtle class of metrics with essential singularities. The former admits a clean intrinsic description; the latter, by construction, concentrates all potential intrinsic obstructions.

2 Setup and definitions

Let $D^2 \subset \mathbb{R}^2$ denote the closed unit disk and write

$$D_*^2 = D^2 \setminus \{0\}$$

for the punctured disk. Throughout, we consider Riemannian metrics of low regularity on D_*^2 .

Definition 2.1. A C^2 Riemannian metric on D_*^2 is a symmetric, positive-definite $(0,2)$ -tensor field g such that, in any local coordinate chart, the coefficients of g are C^2 functions.

Given such a metric g , we regard (D_*^2, g) as a metric space with distance induced by the Riemannian length of curves. The metric space may or may not be complete.

Definition 2.2. Let (D_*^2, g) be a C^2 Riemannian manifold.

- The *metric completion* of (D_*^2, g) is the Cauchy completion $\overline{(D_*^2, d_g)}$ as a metric space.
- We say that g has a *removable singularity* at 0 if there exists a C^2 Riemannian metric $\tilde{g}$ on all of D^2 such that $\tilde{g}|_{D_*^2} = g$. Equivalently, the completion of (D_*^2, g) can be endowed with a smooth structure making the added point into a C^2 Riemannian point.
- If no such C^2 extension exists, we say that g has an *essential singularity* at 0.

We will describe removable singularities in terms of geodesic polar coordinates and the behavior of small geodesic circles. Assume for the moment that the metric completion of (D_*^2, g) adds a single point, which we again denote by 0.

Definition 2.3. Suppose g is a C^2 Riemannian metric on D_*^2 and that the metric completion adds a single point 0. We say that *geodesic polar coordinates* (r, θ) centered at 0 exist if there is a neighborhood $U \subset D_*^2$ of 0 such that every point in U is joined to 0 by a unique minimizing geodesic, and $U \setminus \{0\}$ can be parametrized by

$$(0, \varepsilon) \times S^1 \rightarrow U \setminus \{0\}, \quad (r, \theta) \mapsto \exp_0(rv(\theta))$$

for some smooth parametrization $v(\theta)$ of the unit tangent directions at 0.

In such coordinates, the metric can be written in the general form

$$g = dr^2 + A(r, \theta)^2 d\theta^2 \quad (2.1)$$

for some positive function $A(r, \theta)$ on $(0, \varepsilon) \times S^1$. The *length of the geodesic circle* of radius r is then

$$L(r) = \int_0^{2\pi} A(r, \theta) d\theta. \quad (2.2)$$

3 Removable and essential C^2 singularities

We now state a simple intrinsic criterion characterizing removable singularities at the puncture.

Theorem 3.1 (Removable singularity criterion). *Let g be a C^2 Riemannian metric on D_*^2 , and suppose the metric completion of (D_*^2, g) adds a single point, denoted again by 0. Assume that geodesic polar coordinates (r, θ) centered at 0 exist for $0 < r < \varepsilon$ and that g takes the form (2.1).*

Then the following are equivalent:

- (i) *g extends to a C^2 Riemannian metric on all of D^2 , i.e. the singularity at 0 is removable.*
- (ii) *The Gaussian curvature K of g is bounded in a punctured neighborhood of 0, and the lengths of geodesic circles satisfy*

$$L(r) = 2\pi r + O(r^3) \quad \text{as } r \rightarrow 0.$$

- (iii) *In geodesic polar coordinates (r, θ) , the metric admits the C^2 expansion*

$$g = dr^2 + r^2 d\theta^2 + O(r^3)$$

in the sense that $A(r, \theta) = r + O(r^3)$ with two continuous derivatives in (r, θ) up to $r = 0$.

Remark 3.2. Condition (iii) says that, in geodesic polar coordinates, g agrees to second order with the Euclidean metric on $\mathbb{R}^2$ at the origin. Equivalently, after passing to local Cartesian coordinates via $x = r \cos \theta$, $y = r \sin \theta$, the metric coefficients $g_{ij}(x, y)$ extend as C^2 functions with $g_{ij}(0) = \delta_{ij}$.

Sketch of proof. (i) $\Rightarrow$ (ii): If g extends as a C^2 Riemannian metric to D^2 , then in normal coordinates centered at 0 the standard expansion for C^2 metrics yields

$$g_{ij}(x) = \delta_{ij} + O(|x|^2),$$

with bounded curvature and Christoffel symbols. Passing to polar coordinates, one obtains $g = dr^2 + r^2 d\theta^2 + O(r^3)$ and $L(r) = 2\pi r + O(r^3)$, while the curvature remains bounded near the origin; see, for example, standard treatments of normal coordinates for C^2 metrics.

(ii) $\Rightarrow$ (iii): Bounded curvature together with the Euclidean first-order behavior of $L(r)$ implies that there is no conical angle defect or excess at 0, and rules out cusp or horn-type

degeneracies in the radial direction. One constructs geodesic polar coordinates and applies the Gauss–Bonnet and Jacobi field estimates to show that $A(r, \theta)$ admits an expansion

$$A(r, \theta) = r + O(r^3)$$

with two continuous derivatives in (r, θ) up to $r = 0$. The key ingredient is that bounded curvature controls the behavior of radial geodesics and their variation in θ , while the $L(r)$ expansion enforces the correct angular measure at first order. Together these give the claimed C^2 control.

(iii) $\Rightarrow$ (i): Assuming $A(r, \theta) = r + O(r^3)$ with C^2 control, we can pass to local Cartesian coordinates $x = r \cos \theta$, $y = r \sin \theta$. A direct computation shows that the metric coefficients $g_{ij}(x, y)$ extend continuously with two continuous derivatives to $(x, y) = (0, 0)$ and satisfy $g_{ij}(0, 0) = \delta_{ij}$. Thus g extends to a C^2 Riemannian metric on a neighborhood of 0 in D^2 . Gluing with g on D_*^2 defines a C^2 metric on all of D^2 . This shows that the singularity is removable.

Full details can be supplied by combining standard normal-coordinate expansions with straightforward coordinate computations; we omit them here. $\square$

The theorem leads to the following intrinsic definition.

Definition 3.3. Let g be a C^2 Riemannian metric on D_*^2 whose metric completion adds a single point, and assume geodesic polar coordinates exist near that point. We say that:

- the singularity at 0 is *removable* if the equivalent conditions of Theorem 3.1 hold;
- otherwise, the singularity is *essential*. In this case, at least one of the following must occur:
 - unbounded curvature near 0;
 - non-Euclidean collapse of geodesic circles, $L(r) \neq 2\pi r + O(r^3)$;
 - failure of the C^2 polar expansion $g = dr^2 + r^2 d\theta^2 + O(r^3)$.

In particular, essential singularities encapsulate cone-type behavior (angle defect or excess), cusp and horn ends, and more exotic oscillatory degeneracies that cannot be removed by any C^2 change of coordinates.

4 Implications for the C^2 isometric immersion problem

We now explain how the removable/essential dichotomy yields a structural reduction of the C^2 isometric immersion problem.

Let (M, g) be a two-dimensional C^2 Riemannian manifold which may be incomplete. Consider the set of ends of M which are metrically incomplete in the sense that some geodesics exit the manifold in finite length. Around any such end, one may choose a coordinate chart identifying a neighborhood with a punctured disk D_*^2 and apply the preceding analysis.

Definition 4.1. We say that (M, g) has *only removable singularities* if, in every such punctured disk chart, the corresponding singularity is removable in the sense of Theorem 3.1. We say that (M, g) admits an *essential singularity* if there is at least one punctured end whose singularity is essential.

Proposition 4.2. *Let (M, g) be a two-dimensional C^2 Riemannian manifold.*

1. *If (M, g) has only removable singularities, then its metric completion $(\overline{M}, d_g)$ carries a natural structure of a C^2 Riemannian surface $\overline{M}$ extending g . In this case, there is no intrinsic obstruction to the existence of a local C^2 isometric immersion of $(\overline{M}, \overline{g})$ into $\mathbb{R}^3$; any obstruction must arise from extrinsic compatibility conditions (Gauss–Codazzi) or global topological issues.*
2. *Conversely, any intrinsic C^2 -level obstruction to isometric immersion of (M, g) into $\mathbb{R}^3$ must be concentrated at essential singularities.*

Idea of proof. If all singularities are removable, Theorem 3.1 allows one to extend the metric across each incomplete end to obtain a complete C^2 Riemannian surface $(\overline{M}, \overline{g})$. Intrinsically, $\overline{M}$ is no different from a C^2 manifold without singularities; in particular, the only remaining constraints for local isometric immersion are the usual compatibility conditions between the induced metric, second fundamental form, and curvature encoded in the Gauss–Codazzi equations. In this sense, the intrinsic metric structure poses no additional obstruction.

If, on the other hand, (M, g) admits an essential singularity, then by definition at least one of the intrinsic conditions in Theorem 3.1 fails in any punctured chart around that end. This failure reflects a genuine geometric defect (such as curvature blow-up or non-Euclidean angle collapse) which cannot be eliminated by coordinate changes. Any attempted extension of an isometric immersion across such an end would have to represent this defect extrinsically, and it is natural to view such ends as the only places where truly intrinsic obstructions to C^2 immersion can arise. $\square$

Thus, at the level of intrinsic geometry, the C^2 isometric immersion problem naturally splits into two regimes:

- a large, geometrically natural class of metrics with only removable singularities, for which the intrinsic metric structure is not the source of difficulty; and
- a small but subtle class of metrics with essential singularities, which concentrate all possible intrinsic obstructions.

5 Discussion: relation to the Weyl problem

The classical Weyl problem asks whether a given C^2 Riemannian metric of positive Gauss curvature on the two-sphere S^2 can be realized as the induced metric on a convex surface in $\mathbb{R}^3$. This has been answered affirmatively under various regularity assumptions by work of Nirenberg, Pogorelov, Guan–Li, and others.

From the point of view of the present note, the Weyl problem operates in an intrinsically very special regime:

- the underlying manifold $M = S^2$ is compact and without boundary;
- the metric g is globally C^2 and positive definite;
- there are *no* incomplete ends or singularities of any kind.

In particular, any metric in the Weyl class has only removable singularities in the sense of our definitions (indeed, none at all), with bounded curvature and Euclidean local behavior of small geodesic circles. Thus the Weyl class lies entirely in the “intrinsically safe” regime of Proposition 4.2: it introduces no intrinsic obstructions to C^2 isometric immersion.

The removable/essential decomposition therefore clarifies the role of Weyl-type results in the broader C^2 immersion problem. While the Weyl problem and its solutions are profound and highly nontrivial, they address a class of metrics for which the intrinsic geometry is already regular, and hence do not interact with the essential singularities identified here. In particular, attempts to extend or adapt Weyl-style methods—which rely heavily on convexity, positivity of curvature, and global regularity—cannot hope to resolve the remaining C^2 immersion cases, which lie wholly within the essential-singularity regime.

From this perspective, the intrinsic classification developed in this note serves as a structural guide: it singles out the essential singularities as the genuine frontier for future work on low-regularity isometric immersions, and suggests that new techniques, sensitive to intrinsic blow-up and non-Euclidean collapse at ends, will be required. At the same time, it provides a unified way to regard all Weyl-type metrics, and many more, as intrinsically benign from the immersion standpoint.

Concluding remarks

We have given an intrinsic criterion for when a C^2 metric on a punctured disk admits a smooth completion at the puncture, and used this to decompose C^2 singularities into removable and essential types. This yields a reduction of the C^2 isometric immersion problem: metrics with only removable singularities admit no intrinsic obstructions to immersion in $\mathbb{R}^3$, while essential singularities concentrate all potential intrinsic difficulty.

Even in this basic form, the removable/essential dichotomy offers a conceptual reorganization of the low-regularity immersion problem and pinpoints where new geometric ideas are needed. It would be interesting to refine the analysis of essential singularities, to develop model examples and classification results, and to study how various extrinsic embedding techniques interact with the intrinsic pathologies isolated here.

References

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